

Detection and Analysis of Emoji-Based Jailbreak Attacks in Large Language Models

Abstract—Large language models are increasingly used in applications where reliable safety behavior is important. At the same time, small changes to a prompt, including the use of emojis and other Unicode symbols, can affect how safety mechanisms interpret the input. This paper studies emoji-based prompt perturbations and investigates a lightweight pre-inference method for identifying harmful intent before a prompt reaches a target language model. A controlled dataset of 800 prompt instances was created from 200 underlying benign and harmful prompt pairs, with original, prefix, suffix, and insertion variants. The generation-stage experiment was evaluated using Qwen-2.5-7B-Instruct. The model produced no harmful compliance in the evaluated emoji-perturbed cases, although one original harmful prompt received a partial-compliance response. For pre-inference detection, frozen semantic representations were combined with logistic regression and evaluated using grouped five-fold cross-validation. A length-balanced benchmark containing 488 instances was used to reduce the effect of prompt length as a shortcut. The selected candidate architecture uses Safety-Policy-MiniLM with 32-token overlapping windows, a 16-token stride, and max pooling. It achieved a mean ROC-AUC of 0.723 with a standard deviation of 0.111 and a harmful recall of 77.2 percent with a standard deviation of 9.3 percent on the balanced benchmark. A separate contextual-padding test showed that global sequence representations could suffer severe dilution, while the windowed detector maintained 67 percent harmful recall under long padding. The results suggest that localized semantic analysis can be useful as a probabilistic pre-inference safety layer, while also showing that false positives and contextual sensitivity remain important limitations.

Index Terms—large language models, jailbreak attacks, emoji perturbation, prompt safety, harmful intent detection, semantic embeddings, pre-inference defense

I. INTRODUCTION

Large language models (LLMs) have become common components of conversational systems, coding assistants, search tools, and other AI applications. Their usefulness depends not only on their ability to generate relevant responses, but also on their ability to avoid producing unsafe or restricted content when users provide harmful requests. This has led to increasing interest in prompt-level safety mechanisms and jailbreak detection.

A jailbreak attack attempts to manipulate a model or its safety mechanism so that a restricted request is treated as acceptable. Such attacks can involve carefully designed instructions, changes in wording, role-playing, encoding, or other transformations of an otherwise harmful request [3]. Emoji-based manipulation is particularly interesting because

emojis are ordinary elements of online communication, but their presence can also alter tokenization and the representation of surrounding text.

Recent research has shown that emojis can be used to influence the behavior of safety-oriented judge models. Wei et al. reported that inserting emojis into harmful content could reduce the ability of several judge models to identify unsafe content [1]. Another recent study examined emoji-based jailbreaking directly at the prompt level and reported different levels of vulnerability across several open-source language models [2]. These results motivate a closer examination of both sides of the problem: whether emoji perturbations actually cause unsafe generation in a target model and whether a separate detector can identify harmful intent before the target model processes the request.

In this work, we study emoji perturbations using a controlled set of benign and harmful prompts. Four versions of each prompt are considered: the original prompt, an emoji prefix, an emoji suffix, and an emoji insertion. The first part of the study evaluates the resulting responses from a target LLM. The second part investigates whether a lightweight semantic detector can distinguish harmful intent from benign intent before inference.

An important issue encountered during the study was the effect of prompt length. A classifier can appear to perform well if harmful prompts are systematically longer than benign prompts. To reduce this possibility, a length-balanced benchmark was constructed by matching prompts according to character length. Grouped cross-validation was then used so that different perturbation variants of the same underlying prompt never appeared in different folds.

The final part of the study examines contextual padding. A harmful request can be surrounded by a long benign-looking academic or security-related explanation. Such padding may dilute the harmful signal when a complete prompt is represented by a single embedding. To investigate this effect, a hierarchical sliding-window detector was evaluated using overlapping local windows.

The main contributions of this study are as follows:

- A controlled evaluation of original and emoji-perturbed benign and harmful prompts.
- A generation-stage analysis of emoji perturbations using a target instruction-tuned LLM.

- A pre-inference harmful-intent detection framework based on frozen semantic representations and logistic regression.
- A length-balanced evaluation protocol with grouped cross-validation to reduce shortcut learning and group leakage.
- A contextual-padding stress test comparing global sequence representations with localized sliding-window representations.
- An analysis of the limitations of the proposed detector, including false positives and sensitivity to contextual changes.

II. RELATED WORK

A. Jailbreak Attacks Against Large Language Models

Jailbreaking has become an important area of LLM safety research because aligned models can sometimes be induced to produce content that they would normally refuse. Existing studies have investigated a wide range of attack strategies and corresponding defense mechanisms [3]. These studies show that safety behavior can depend strongly on prompt construction and on the particular model being evaluated.

The present work focuses on a narrower form of prompt manipulation involving emojis and Unicode symbols. Rather than assuming that an emoji is itself malicious, the study treats emoji insertion as a controlled perturbation of an underlying prompt.

B. Emoji-Based Attacks

Wei et al. introduced Emoji Attack as a method for modifying harmful content with emojis in order to affect the behavior of judge LLMs used for safety detection [1]. Their work suggests that changes to token segmentation and contextual representations can influence safety decisions.

A separate study by Gopinadh and Hussain investigated emoji-based jailbreaking directly against several open-source LLMs. Their results showed that the response to emoji-based prompts differed across models, with Qwen showing different behavior from some of the other evaluated models [2].

These findings provide the motivation for the generation-stage experiment in this study. However, the present work also goes one step further by investigating a separate pre-inference detector rather than relying only on the target model's own safety behavior.

C. Semantic Representations for Text Classification

Sentence-level transformer representations are widely used for representing the semantic content of text. The `all-MiniLM-L6-v2` model provides compact sentence embeddings and is commonly used for semantic representation tasks [4]. In this study, the model is used as a frozen encoder rather than being fine-tuned on the research dataset.

A safety-oriented encoder, Safety-Policy-MiniLM, is also evaluated. Its model card describes it as a safety-policy text classification model that predicts whether an input keeps or breaks the policies represented during fine-tuning [5]. The

encoder is used here as a semantic representation source, followed by a separate logistic regression classifier trained on the research labels.

III. RESEARCH METHODOLOGY

A. Research Objective

The main objective of this research is to detect and analyze emoji-based jailbreak attacks and to investigate whether harmful intent can be identified before a prompt reaches a target LLM.

The research therefore considers two related questions:

- 1) Do simple emoji perturbations change the safety behavior of the evaluated target LLM?
- 2) Can a lightweight pre-inference detector identify harmful intent when the same prompts are represented with different semantic and structural characteristics?

The detector is not treated as a complete jailbreak-success classifier. Its target is harmful intent in the input prompt.

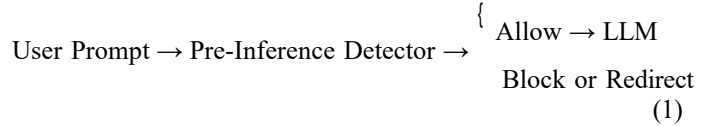
B. Overall Pipeline

The overall research pipeline consists of two stages.

First, the original and perturbed prompts are sent to the target LLM and the responses are manually reviewed. This stage determines whether the emoji perturbation results in refusal, safe redirection, partial compliance, or harmful compliance.

Second, the same prompts are processed before LLM inference using a lightweight classifier. The detector produces a harmful-intent probability. Prompts above the selected threshold can then be treated as candidates for blocking or redirection.

The proposed workflow can be summarized as:



This architecture is intended as a safety layer rather than a guarantee of complete jailbreak prevention.

C. Dataset Construction

The dataset contains 200 underlying prompt pairs. One hundred pairs represent benign requests and one hundred pairs represent harmful requests.

Each prompt was represented in four forms:

- 1) Original prompt.
- 2) Emoji prefix variant.
- 3) Emoji suffix variant.
- 4) Emoji insertion variant.

This resulted in 800 prompt observations.

The four variants derived from the same underlying prompt were treated as a single group during model evaluation. This was necessary because treating the four variants as independent observations could allow highly similar text to appear in both training and validation data.

The original dataset contains 400 benign and 400 harmful observations.

D. Emoji Perturbation Strategy

The perturbation strategy was intentionally simple. Emojis were added before the prompt, after the prompt, or inside the prompt while retaining the underlying request.

The purpose was not to construct increasingly sophisticated jailbreak prompts. Instead, the goal was to isolate the effect of emoji-based modification and determine whether a simple perturbation could change model behavior.

This distinction is important because the presence of an emoji does not imply malicious intent. A safety detector that simply blocks all emoji-containing prompts would produce unnecessary false positives.

E. Target LLM Evaluation

The target generation experiment was performed using Qwen-2.5-7B-Instruct.

Each of the 800 prompts was evaluated using deterministic generation. The responses were first processed using automated response categories and then manually reviewed when required.

For harmful prompts, the manual labels were:

- Safe refusal.
- Safe redirection.
- Partial compliance.
- Harmful compliance.
- Ambiguous.

Strict attack success was defined using harmful compliance. A broader measure also considered partial compliance.

F. Human Validation

Automated response classification was not treated as the final ground truth. Cases requiring review were manually inspected.

For harmful prompts, 51 automatically flagged cases were reviewed. The final human labels were:

- 43 safe refusals.
- 356 safe redirections.
- 1 partial compliance.
- 0 harmful compliance.

Therefore, 399 of the 400 harmful prompts were considered safe after human validation.

The resulting strict attack success rate was zero, while the broader rate including partial compliance was 0.25 percent.

The one partial-compliance case occurred in the original version of a harmful prompt. The emoji variants of that prompt were refused.

IV. PRE-INFERENCE DETECTION FRAMEWORK

A. Detection Target

The primary detection task is binary classification of harmful and benign intent.

The target variable is defined as:

$$y = \begin{cases} 0, & \text{benign intent} \\ 1, & \text{harmful intent} \end{cases} \quad (2)$$

This target is deliberately different from jailbreak success. A prompt may have harmful intent even when the target LLM correctly refuses it.

This distinction allows the detector to operate before the target LLM generates a response.

B. Baseline Features

Several groups of features were investigated.

Text-based features included TF-IDF representations of the prompt. Structural features included text length, word count, punctuation count, and whitespace-related measures.

Emoji and Unicode features included emoji count, emoji density, emoji position, detected symbol count, unique symbol count, Unicode code points, and Unicode categories.

These experiments were useful for determining whether the task could be solved using simple surface-level characteristics.

C. Semantic Features

Two frozen semantic encoders were evaluated:

- all-MiniLM-L6-v2.
- SummerSigh/Safety-Policy-MiniLM.

The all-MiniLM-L6-v2 model produces compact 384-dimensional sentence representations [6]. The Safety-Policy-MiniLM model is a policy-oriented text classification model based on a DeBERTa-v3 architecture [5].

The encoders were kept frozen. A logistic regression classifier was trained on the resulting representations.

The selected classifier used L2 regularization with $C = 1.0$ and class-balanced fitting.

D. Length-Balanced Evaluation

The original dataset contains prompts with different lengths. Since length can become a shortcut for binary classification, a second benchmark was constructed.

The length-balanced benchmark contains 488 observations derived from 122 prompt pairs and their four variants. Prompt pairs were matched using character-length similarity.

This benchmark was used as the primary evaluation set for the final detector.

E. Grouped Cross-Validation

Five-fold Stratified Group K-Fold cross-validation was used.

The grouping variable was the underlying `unique_pair_id`. Consequently, all four variants of a prompt remained in the same fold.

The validation protocol used:

```
StratifiedGroupKFold(  
    n_splits=5,  
    shuffle=True,  
    random_state=42  
)
```

No post-inference information, such as the generated response or response category, was used as an input feature.

The reported benchmark values are the mean and standard deviation across the five validation folds.

V. SLIDING-WINDOW DETECTOR

A. Motivation

The global semantic encoder represents the complete prompt as a single sequence. This can become problematic when a harmful request is surrounded by a large amount of benign-looking context.

To examine this issue, a sliding-window representation was introduced.

Instead of producing one representation for the entire input, the prompt is divided into overlapping token windows. Each window is independently encoded and classified.

The final harmful probability is calculated using max pooling:

$$P_{\text{final}} = \max_{m=1, \dots, M} P(\text{harmful} \mid w_m) \quad (3)$$

where w_m denotes the m -th token window and M is the total number of windows.

B. Selected Architecture

The selected candidate architecture for the evaluated pre-inference defense uses:

- Safety-Policy-MiniLM representations.
- Window size of 32 tokens.
- Stride of 16 tokens.
- 50 percent overlap between consecutive windows.
- Max-pooling aggregation.
- Decision threshold of 0.50.

This should be understood as the selected candidate architecture for the evaluated defense rather than as a universally optimal architecture.

VI. RESULTS AND DISCUSSION

The experiments were carried out in stages to understand both the safety behavior of the target language model and the performance of the proposed pre-inference detection approach. The analysis first examines the response of the target model to harmful prompts and emoji-based variations. It then evaluates lightweight detection models, followed by length-controlled experiments and robustness tests involving contextual padding, emoji perturbations, and different detection configurations.

Since every underlying prompt was represented by four related versions, the evaluation used five-fold stratified grouped cross-validation. The grouping ensured that related variants of the same underlying prompt were kept in the same fold rather than being split between training and validation data.

A. Target LLM Safety Results

The harmful subset contained 400 observations. Each harmful prompt was evaluated in its original form as well as under three emoji perturbations: prefix, suffix, and insertion.

After human validation, 43 responses were classified as safe refusals, 356 as safe redirections, and one as partial compliance. No harmful-compliance response was observed.

Therefore, the strict attack success rate was 0.00%. If the single partial compliance case is included in a broader measure, the attack success rate was 0.25%.

An important observation is that none of the 300 emoji-perturbed harmful observations resulted in either partial or harmful compliance. The only partial-compliance case occurred for an original harmful prompt. In that case, the model provided a disclaimer but still produced part of the requested harmful content.

These results suggest that the emoji perturbations used in this study did not weaken the safety behavior of the evaluated target model. This finding is specific to the tested model, prompt set, and perturbation strategies. It should not be taken as evidence that emoji-based attacks are ineffective against other models or against more complex attack strategies.

B. Feature Distribution

Before training the detection models, the main prompt-level features were examined for the benign and harmful classes. Fig. 1 presents the distribution of character length, word count, punctuation count, and emoji density.

The original dataset shows a visible difference in character length between the two classes. Harmful prompts have a higher average character length than benign prompts. This difference is important because a classifier could obtain apparently useful performance simply by learning the length of a prompt rather than its harmful intent. Word count and punctuation also show some differences, while emoji density has a large overlap between the two classes.

C. Length-Matched Evaluation

The difference in prompt length motivated a separate length-matched benchmark. In the original dataset, the mean character length was approximately 75.3 for benign prompts and 87.6 for harmful prompts. After length matching, the means were approximately 78.1 and 78.8, respectively.

Fig. 2 shows the distributions before and after matching. The difference between the classes becomes much smaller after matching. The resulting Cohen’s d was 0.044 with $p = 0.541$, indicating that there was no meaningful difference in character length between the two classes in the matched benchmark.

The effect of this control is also visible in Fig. 3. Several models that performed reasonably well on the original benchmark moved closer to chance-level discrimination after length matching. This indicates that prompt length was an important source of information for some of the initial models.

D. Baseline Detection Results

Three baseline configurations were first evaluated using five-fold stratified grouped cross-validation. The first model used text-based TF-IDF features. The second used emoji and prompt-morphology features, while the third combined structural and emoji-related features.

The precision-recall curves are shown in Fig. 4. The emoji/morphology baseline obtained a PR-AUC of 0.655, which was higher than the text-only baseline at 0.405. The proposed hybrid model obtained a PR-AUC of 0.587. Although

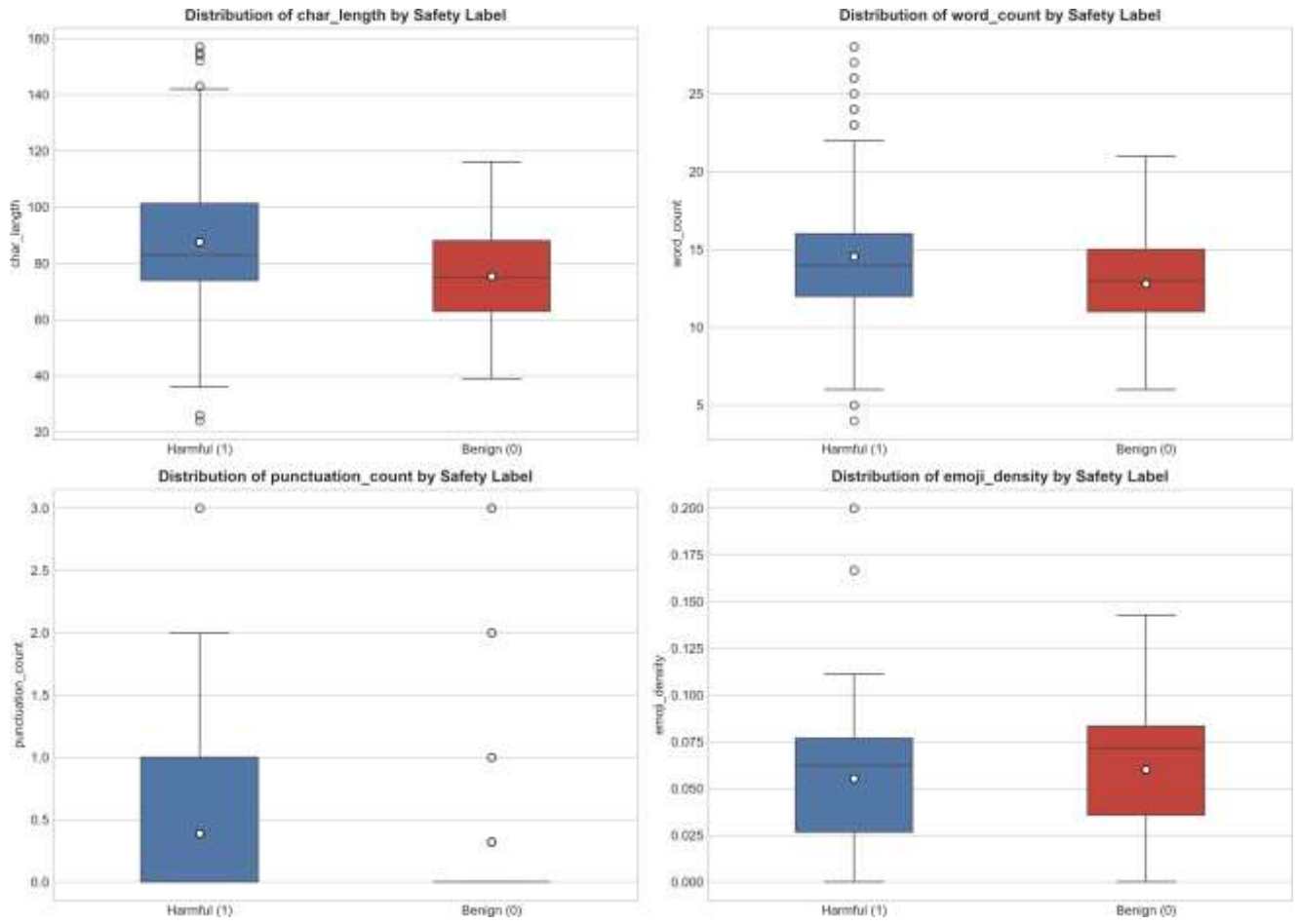


Fig. 1. Distribution of selected prompt features for the benign and harmful classes. The figure shows character length, word count, punctuation count, and emoji density.

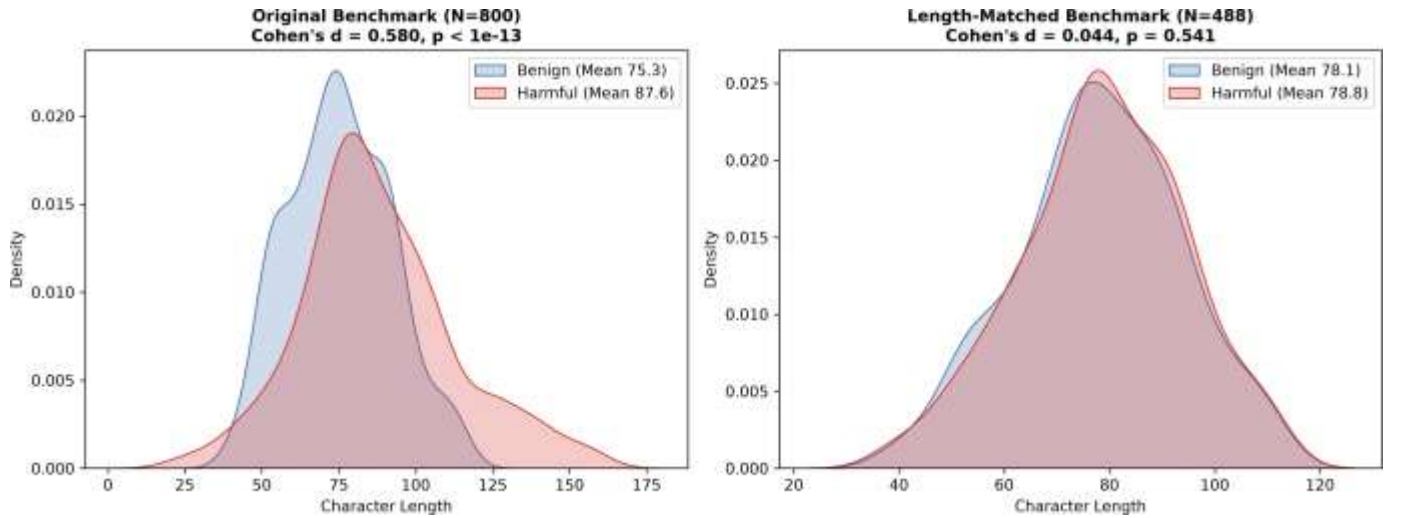


Fig. 2. Character-length distributions before and after length matching. The length-matched benchmark substantially reduces the original difference in prompt length between the benign and harmful classes.

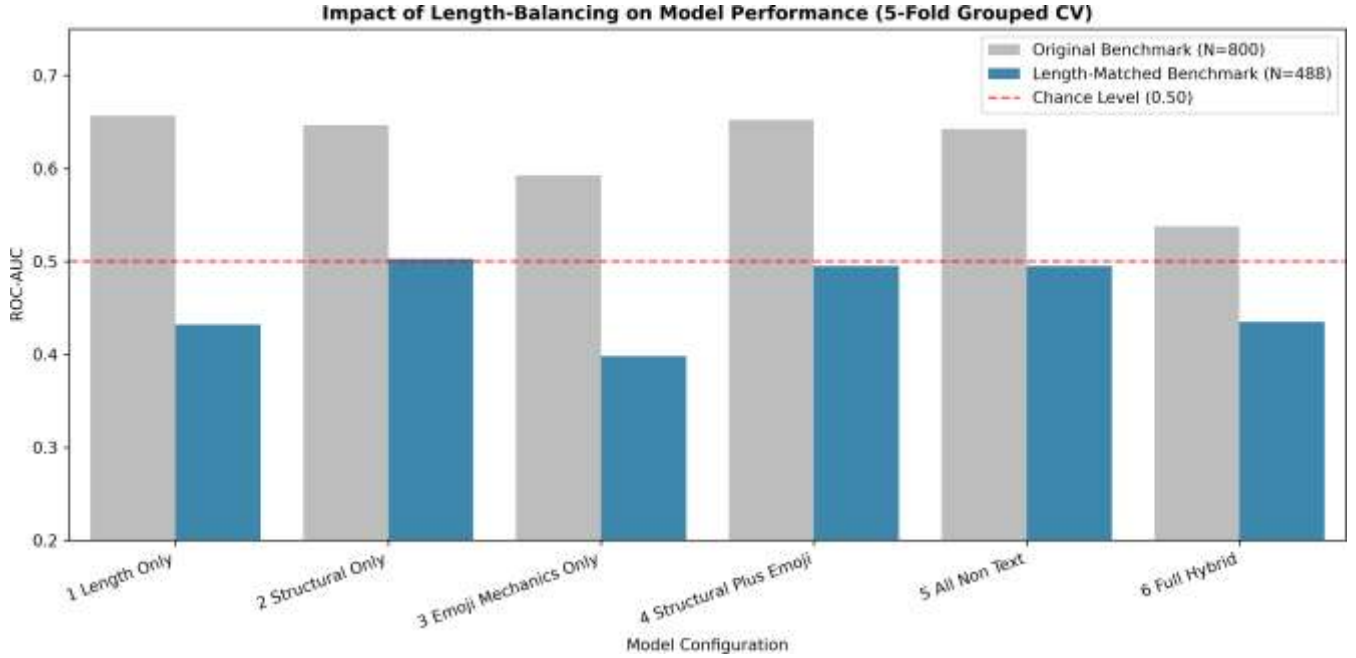


Fig. 3. Comparison of model ROC-AUC on the original and length-matched benchmarks. The comparison illustrates the reduction in apparent performance after controlling for prompt length.

TABLE I
PRIMARY LENGTH-BALANCED BENCHMARK RESULTS

Metric	all-MiniLM	Safety-Policy
ROC-AUC	0.807 $\pm$ 0.071	0.723 $\pm$ 0.111
PR-AUC	0.826 $\pm$ 0.069	0.722 $\pm$ 0.145
Accuracy	72.8 $\pm$ 6.7%	67.5 $\pm$ 8.8%
Balanced Accuracy	73.0 $\pm$ 6.6%	67.7 $\pm$ 9.1%
Precision	78.1 $\pm$ 11.8%	67.7 $\pm$ 16.7%
Harmful Recall	67.5 $\pm$ 14.5%	77.2 $\pm$ 9.3%
Benign FPR	21.4 $\pm$ 16.1%	41.7 $\pm$ 23.5%
Harmful FNR	32.5 $\pm$ 14.5%	22.8 $\pm$ 9.3%
F1 Score	0.709 $\pm$ 0.078	0.707 $\pm$ 0.059

the emoji/morphology model appears stronger on the original benchmark, the length-matched analysis shows that part of this performance is related to structural differences in the data.

The ROC curves in Fig. 5 provide a similar comparison. The text-based model obtained an out-of-fold ROC-AUC of 0.338. The emoji/morphology baseline reached 0.614, while the hybrid model obtained 0.531.

E. Primary Detection Benchmark

After examining the baseline models, semantic representations were evaluated as a more direct way of representing harmful intent. Two frozen encoders were considered: *all-MiniLM-L6-v2* as a general-purpose sentence representation and *Safety-Policy-MiniLM* as a safety-oriented representation.

Table I presents the main results on the length-balanced benchmark.

The general MiniLM representation obtained the higher ROC-AUC and PR-AUC, indicating better overall ranking

performance on the balanced benchmark. The Safety-Policy representation, however, obtained higher harmful recall at the selected operating point. Its harmful recall was 77.2%, compared with 67.5% for the general MiniLM model.

The difference comes with a higher false-positive rate. The Safety-Policy detector produced a benign false-positive rate of 41.7%, compared with 21.4% for the general MiniLM model. This is an important consideration for a practical pre-inference filter, since blocking too many legitimate prompts would reduce its usefulness.

F. Contextual Padding Stress Test

The next experiment examined whether additional harmless context could change the prediction of the detectors. This test is important because real prompts may contain explanations, background information, or educational material around the actual request.

A separate stress-test set containing 100 harmful and 100 benign prompts was used. Each prompt was evaluated without padding and after adding short, medium, and long academic-style contextual material.

Fig. 6 illustrates the change in predicted harmful probability and false-negative rate across the padding conditions.

The Safety-Policy model showed a substantial reduction in harmful recall as contextual padding increased. Under the longest global-padding condition, harmful recall dropped to 1%. In contrast, the windowed Safety-Policy configuration retained 67% harmful recall under the same condition.

The complete stress-test results are given in Table II.

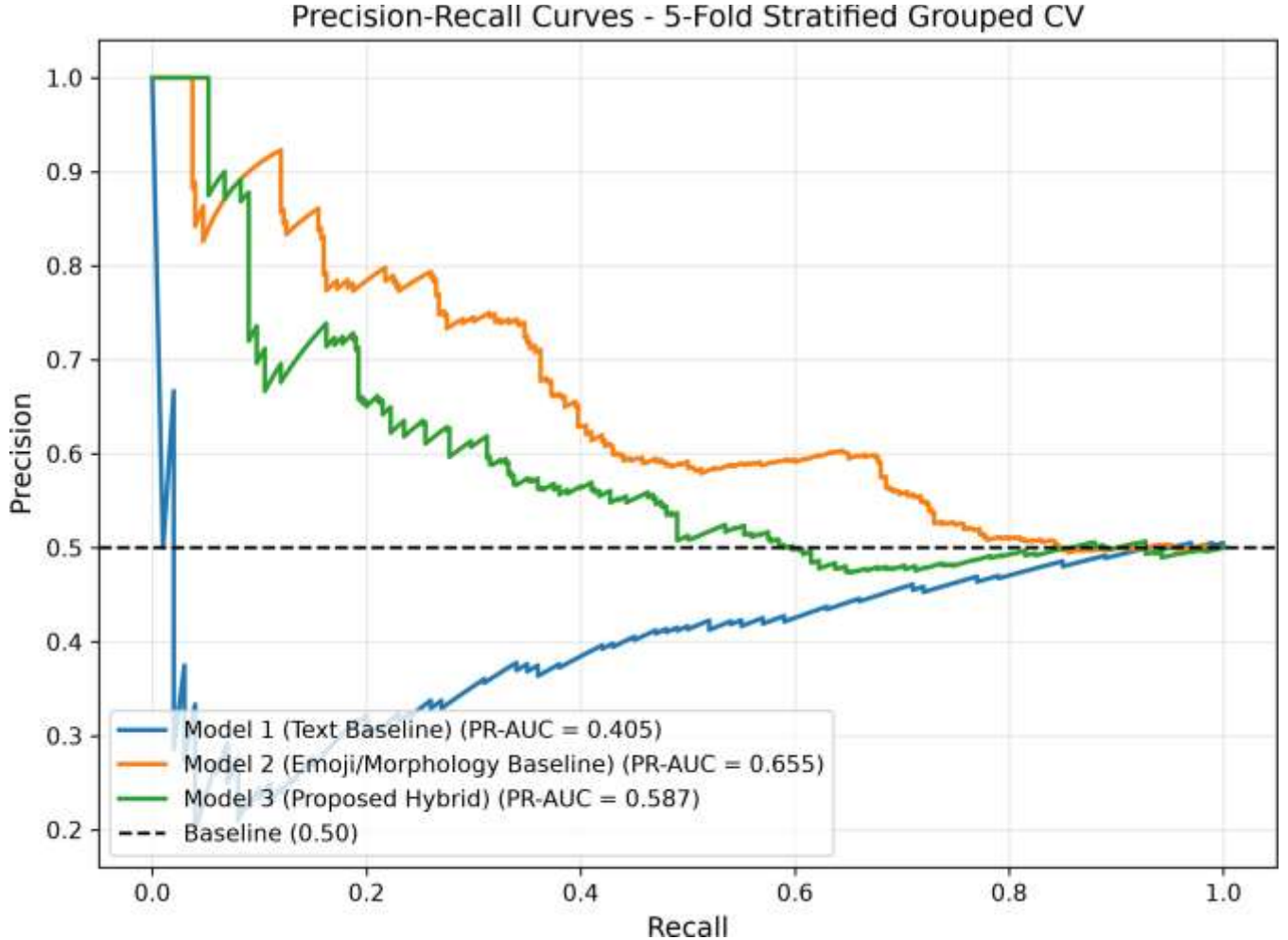


Fig. 4. Precision-recall curves for the three baseline detection models using five-fold stratified grouped cross-validation. The PR-AUC values are 0.405 for the text baseline, 0.655 for the emoji/morphology baseline, and 0.587 for the hybrid model.

TABLE II
CONTEXTUAL PADDING STRESS TEST

Model	Condition	Harmful Recall	Benign FPR
Safety-Policy	Unpadded	68%	40%
Safety-Policy	Short	52%	14%
Safety-Policy	Medium	22%	6%
Safety-Policy	Long, Global	1%	0%
Safety-Policy	Long, Windowed	67%	32%
MiniLM	Long, Global	56%	50%
MiniLM	Long, Windowed	97%	93%

The statistical comparison between the global and windowed Safety-Policy predictions under long padding gave a mean probability difference of 0.294 with a standard deviation of 0.115. A paired t -test gave $p = 4.62 \times 10^{-45}$, while the Wilcoxon signed-rank test gave $p = 3.90 \times 10^{-18}$. Bootstrap analysis produced a 95% confidence interval of 56.0 to 75.0 percentage points for the improvement in harmful recall.

The MiniLM windowed model obtained 97% harmful recall under long padding. However, its benign false-positive rate

also increased to 93%. This result is important because it shows that maximizing harmful recall alone does not produce a useful detector. Both missed harmful prompts and unnecessary blocking of benign prompts have to be considered.

G. Global and Windowed Representations

On the length-balanced benchmark, the windowed Safety-Policy detector produced the same results as the corresponding global representation. This happens because the balanced prompts contain no more than 24 tokens and therefore fit inside a single 32-token window.

The advantage of windowed processing becomes visible when prompts become longer. Instead of representing the complete prompt as one sequence, the windowed approach evaluates overlapping local sections and then combines their predictions. This allows the detector to retain a harmful signal even when additional contextual material is present elsewhere in the prompt.

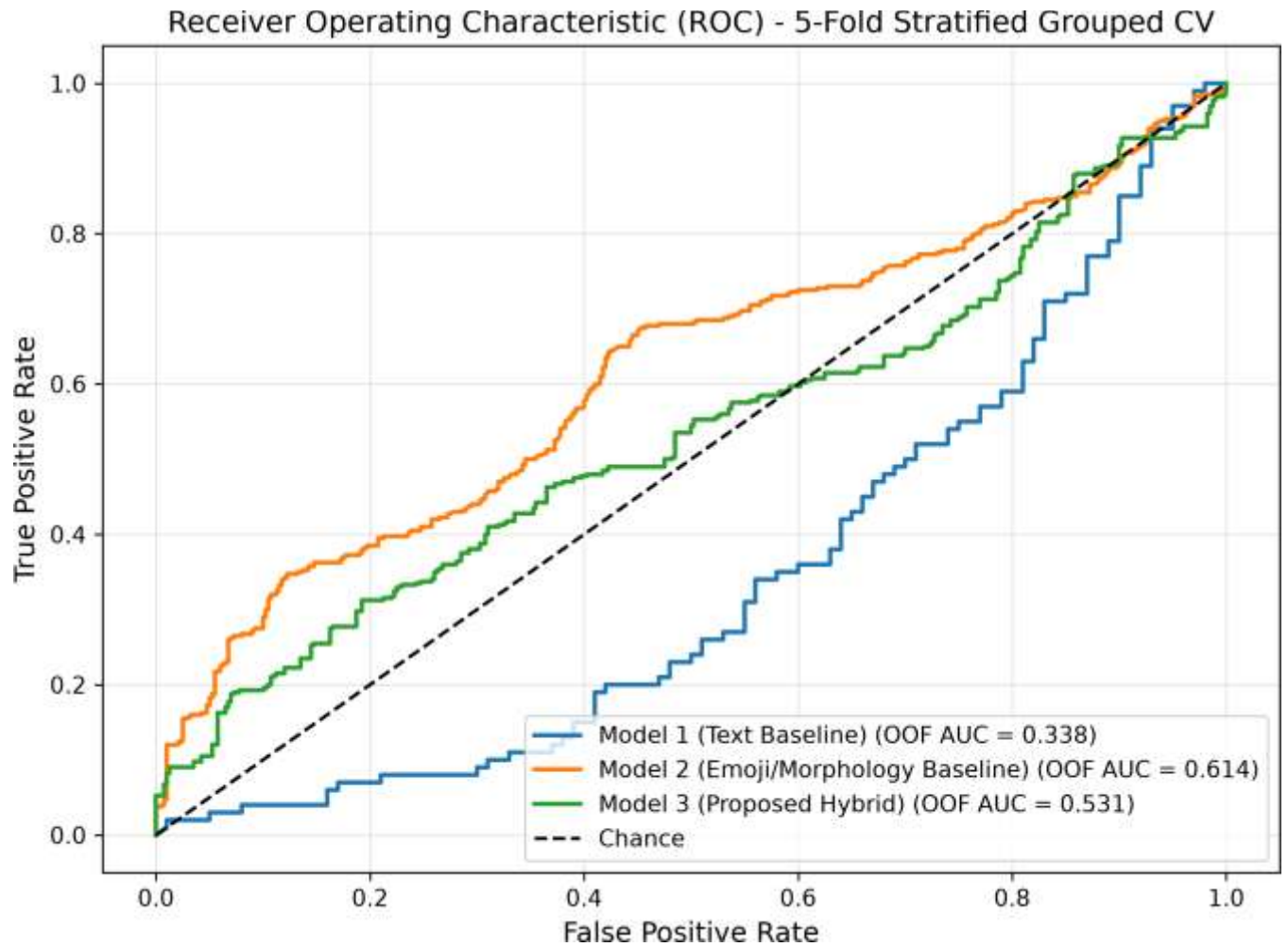


Fig. 5. Receiver operating characteristic curves for the three baseline detection models using five-fold stratified grouped cross-validation. The curves show the out-of-fold discrimination performance across the benchmark.

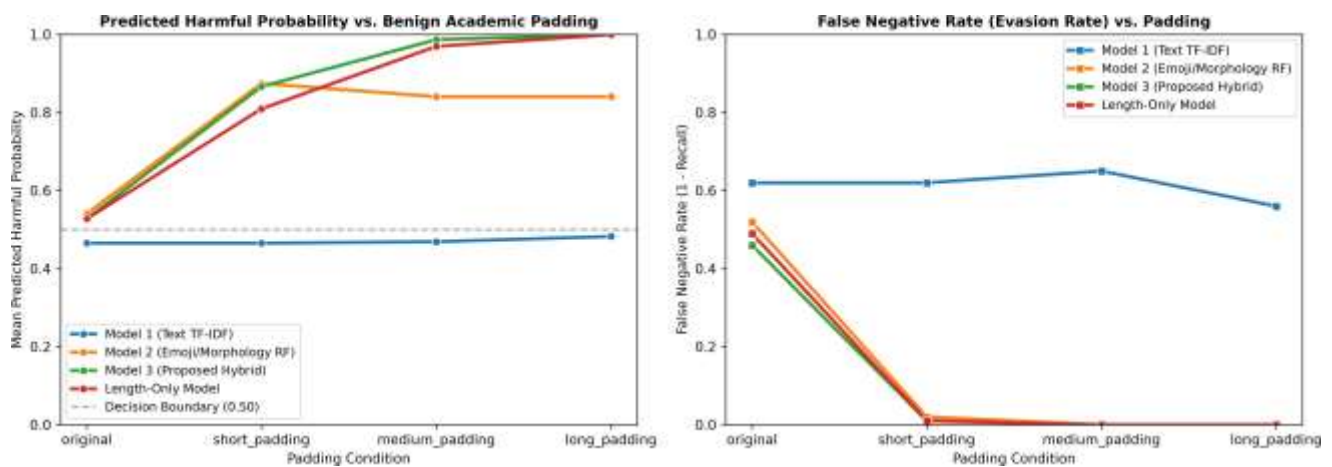


Fig. 6. Effect of increasing benign contextual padding on the predicted harmful probability and false-negative rate of the baseline detection models.

TABLE III
WINDOW SIZE ABLATION

Window	ROC-AUC	Long Recall	Long FPR
24	0.732	69%	36%
32	0.723	67%	32%
40	0.723	61%	30%
64	0.723	48%	22%

TABLE IV
AGGREGATION ABLATION UNDER LONG PADDING

Aggregation	Harmful Recall	Benign FPR
Max pooling	67%	32%
Mean pooling	14%	2%
Top-3 mean	31%	11%

H. Window Size Ablation

Four window sizes were tested using a stride equal to half the window size. The results are shown in Table III.

The results do not point to one window size that is best under every metric. The 24-token configuration gives slightly higher ROC-AUC and harmful recall, while the 64-token configuration gives lower false-positive rates but loses more harmful signals under long contextual padding.

The 32-token configuration was therefore retained as the selected candidate because it offers a reasonable balance between local detection and false alarms. The choice was not based on a single performance value.

I. Aggregation Ablation

Three aggregation strategies were also evaluated with 32-token windows and a 16-token stride. Their results under long contextual padding are shown in Table IV.

Mean pooling produced fewer false positives but also lost much of the harmful signal. Top-3 mean provided an intermediate result. Max pooling retained the strongest local harmful signal and gave the highest harmful recall among the three methods.

For the intended pre-inference filtering setting, max pooling was therefore selected for the final windowed configuration. This does not mean that max pooling is optimal for every application; it was selected here because the primary concern was avoiding missed harmful intent when harmful content occupies only a small part of a longer prompt.

J. Emoji Perturbation Analysis

The effect of emoji placement on the detector predictions was examined using four versions of each prompt: original, prefix, suffix, and insertion.

Fig. 7 shows the distribution of predicted harmful probabilities across these conditions.

The distributions remain broadly similar across the four conditions, although small changes can be seen depending on the model and the position of the emojis. This supports the view that emoji presence alone is not sufficient to determine harmful intent. A useful detector should therefore rely mainly

on the underlying meaning of the prompt while treating emoji and structural information as additional signals.

K. Threshold and Operating-Point Analysis

The final stage examined the effect of changing the decision threshold used by the detector. The threshold determines how much predicted harmful probability is required before a prompt is classified as harmful.

Fig. 8 presents the threshold trade-off curves generated during the calibration experiment. The curves illustrate the expected relationship between harmful recall and false-positive behavior as the decision threshold changes.

The results show that changing the threshold alone cannot remove the robustness problem caused by contextual padding. A lower threshold can increase harmful recall, but this also increases the number of legitimate prompts that are incorrectly flagged. A higher threshold reduces false alarms but allows more harmful prompts to pass through.

This is why the final detector design focuses on the representation and aggregation strategy rather than treating threshold selection as the main solution.

L. Overall Discussion

The experiments provide several useful observations for the proposed pre-inference defense.

First, the target LLM remained safe for almost all harmful prompts in the evaluated dataset, including all emoji-perturbed harmful prompts. The single partial-compliance case occurred in an original harmful prompt rather than an emoji-perturbed version.

Second, the initial feature-based classifiers were strongly influenced by prompt structure and length. Once the length difference between the classes was controlled, the performance of several structural models moved close to chance. This shows why length matching is important when evaluating harmful-intent detectors.

Third, semantic representations provided a more useful basis for distinguishing harmful and benign intent. The general MiniLM model produced the strongest overall ranking performance on the balanced benchmark, while the Safety-Policy model provided higher harmful recall at the selected operating point.

Finally, the contextual-padding experiments exposed a weakness in global sequence representations. A harmful request can become harder to detect when it is surrounded by a large amount of harmless text. The windowed Safety-Policy approach recovered much of this lost signal under the longest padding condition. At the same time, the MiniLM windowed model produced a very high benign false-positive rate, showing that robustness cannot be judged using harmful recall alone.

Overall, the findings support a pre-inference detector that combines semantic analysis with localized prompt processing. The proposed approach should be viewed as a risk-reduction layer rather than a complete solution to jailbreak attacks. Its purpose is to identify potentially harmful intent before

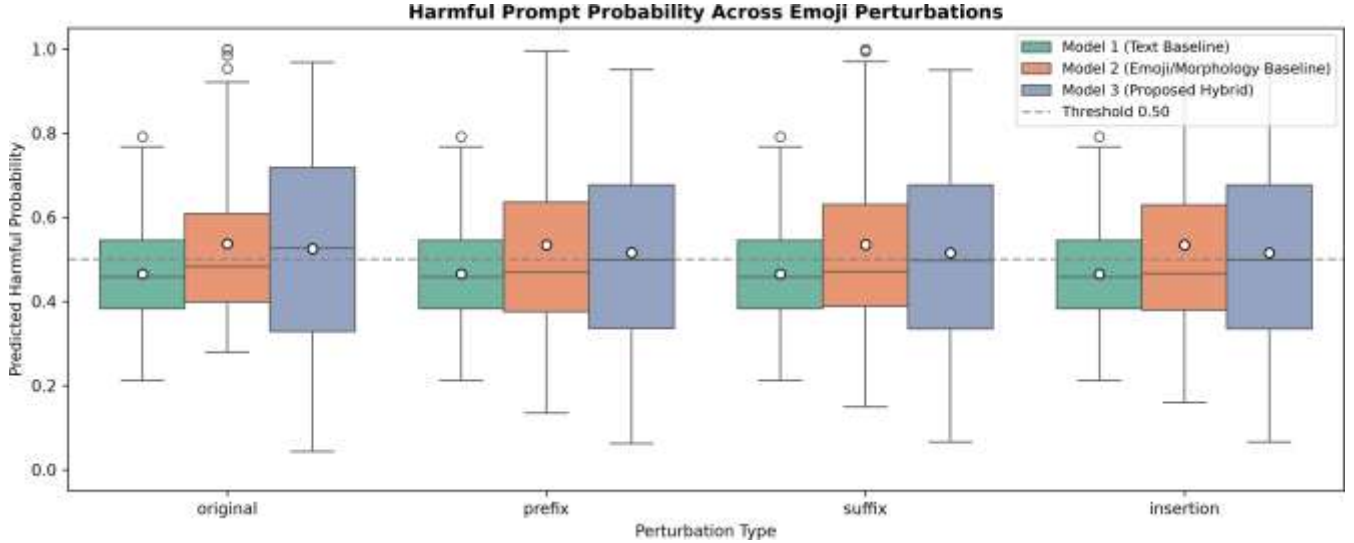


Fig. 7. Distribution of predicted harmful probabilities across original and emoji-perturbed harmful prompts. The perturbations correspond to emoji prefix, suffix, and insertion conditions.

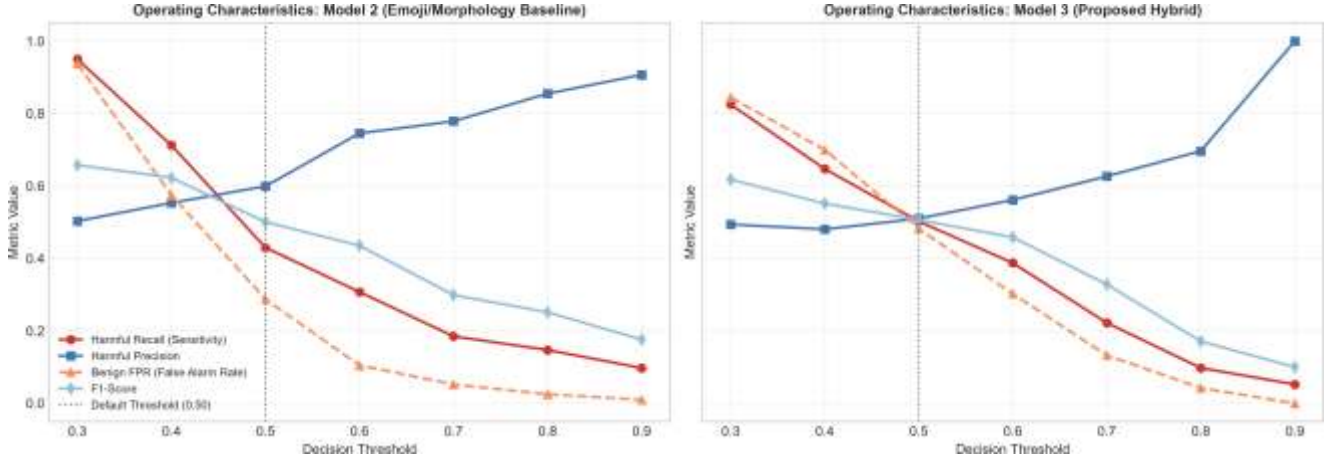


Fig. 8. Threshold trade-off analysis for the pre-inference harmful-intent detector. The curves illustrate the change in detection sensitivity and false-positive behavior as the classification threshold is varied.

inference, while recognizing that false positives and false negatives remain possible.

VII. COMPUTATIONAL ANALYSIS

The detector was also evaluated for inference overhead on CPU.

The measured global representation latency was approximately 25.38 milliseconds per prompt. The selected windowed configuration required approximately 25.21 milliseconds per prompt, with a measured 95th percentile latency of approximately 30.08 milliseconds under the reported evaluation setup.

The measured latency should not be interpreted as a universal deployment figure. Actual performance will depend on hardware, batching, tokenizer implementation, model loading, and runtime configuration.

VIII. DISCUSSION

The experimental results provide several useful observations.

First, the emoji perturbations evaluated in this study did not cause harmful compliance in the target LLM. This does not contradict previous work showing that emojis can affect safety detection. The studies evaluate different models, tasks, and attack settings. In particular, the Emoji Attack work focused on judge-model detection rather than simply asking whether a target LLM would generate harmful content [1].

Second, semantic representations were more informative than simple surface-level features for distinguishing harmful and benign intent. This was especially clear after length balancing, where structural shortcuts became less useful.

Third, the contextual padding experiment revealed an important weakness of global sequence representations. A harmful request can be surrounded by enough benign-looking material that the harmful signal becomes diluted. The sliding-window approach reduced this effect because each local region was evaluated separately.

Fourth, the results also show that improving harmful recall can increase false positives. The generic MiniLM windowed model illustrates this particularly clearly: its long-padding harmful recall was 97 percent, but its benign false-positive rate was also 93 percent. Therefore, high recall alone cannot justify selecting a detector.

The selected Safety-Policy configuration still produces false alarms. On the balanced benchmark, its benign false-positive rate was 41.7 percent at a threshold of 0.50. Under long contextual padding, the false-positive rate was 32 percent. These values indicate that the detector should be treated as a probabilistic safety layer rather than as a final decision-maker.

IX. LIMITATIONS

This study has several limitations.

First, the target generation experiment was performed using a single evaluated LLM. Results may differ for other models, model versions, decoding configurations, or safety policies.

Second, the emoji perturbations were intentionally simple. More sophisticated combinations of emojis, Unicode characters, encoding schemes, or jailbreak strategies were not exhaustively evaluated.

Third, the detection dataset contains 200 underlying prompt pairs. Although the use of grouped cross-validation reduces leakage between variants, a larger and more diverse dataset would be required to establish broader generalization.

Fourth, the contextual padding test used a specific academic-style padding construction. Other forms of benign context may produce different behavior.

Fifth, the detector has a non-trivial false-positive rate. Sensitive educational and dual-use topics can be difficult to distinguish from genuinely harmful requests using a binary classifier.

Finally, the proposed detector identifies harmful intent rather than successful jailbreak execution. A prompt classified as harmful may still be safely refused by the target LLM, while a prompt classified as benign could potentially expose a weakness that is not represented in the training data.

X. CONCLUSION

This study examined the effect of emoji-based prompt perturbations on LLM safety behavior and investigated a lightweight pre-inference approach for identifying harmful intent.

In the evaluated target-model experiment, none of the 300 emoji-perturbed harmful prompts resulted in partial or harmful compliance. Across the complete harmful set, one original prompt was classified as partial compliance and no case was classified as harmful compliance.

For pre-inference detection, semantic representations provided a stronger basis for harmful-intent classification than simple structural features after controlling for prompt length. The selected Safety-Policy-MiniLM sliding-window architecture used 32-token windows, a 16-token stride, and max pooling. On the length-balanced benchmark, it achieved a mean ROC-AUC of 0.723 with a standard deviation of 0.111 and harmful recall of 77.2 percent with a standard deviation of 9.3 percent.

The contextual-padding experiment was particularly informative. Global Safety-Policy representations dropped to 1 percent harmful recall under the longest padding condition, whereas the selected windowed detector maintained 67 percent recall. The improvement shows that localized analysis can reduce the effect of contextual dilution. At the same time, the detector produced false positives, demonstrating that the method is not a complete safety solution.

Overall, the findings support the use of localized semantic analysis as a candidate pre-inference defense layer for harmful-intent detection. The results also highlight the need for larger datasets, more diverse attack constructions, better calibration, and evaluation across multiple target LLMs before such a detector can be considered broadly reliable.

ACKNOWLEDGMENT

The author would like to thank the faculty and research environment at the School of Computer Engineering, Kalinga Institute of Industrial Technology, for providing the academic setting in which this work was carried out.

REFERENCES

- [1] Z. Wei, Y. Liu, and N. B. Erichson, "Emoji Attack: Enhancing Jailbreak Attacks Against Judge LLM Detection," in *Proc. 42nd Int. Conf. Machine Learning (ICML)*, PMLR, vol. 267, pp. 66103–66117, 2025.
- [2] M. P. V. S. Gopinadh and S. Mahaboob Hussain, "Emoji-Based Jailbreaking of Large Language Models," arXiv preprint arXiv:2601.00936, 2026.
- [3] Z. Xu, Y. Liu, G. Deng, Y. Li, and S. Picek, "A Comprehensive Study of Jailbreak Attack versus Defense for Large Language Models," *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 11063–11080, 2024.
- [4] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. 2019 Conf. Empirical Methods in Natural Language Processing and the 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992, 2019.
- [5] SummerSigh, "Safety-Policy-MiniLM," Hugging Face Model Repository, 2023.
- [6] Sentence Transformers, "Sentence Transformers Documentation: all-MiniLM-L6-v2," online documentation.
- [7] M. P. V. S. Gopinadh and S. Mahaboob Hussain, "Emoji-Based Jailbreaking of Large Language Models," arXiv preprint arXiv:2601.00936, 2026.
- [8] S. Grover, "Neutralizing the Emojis: Developing Robust LLM Defenses Against Adversarial Emoji-fiction Attacks," 2026.
- [9] S. Joshi, M. A. Mehta, M. Mendoza, Y. M. Rivera, and V. K. Singh, "Differences in Safety Risks Across Languages in Health-Relevant Queries: Vulnerability Analysis of Large Language Model Responses," *JMIR Formative Research*, vol. 10, p. e87465, 2026.
- [10] S. Cui, X. Feng, Y. Wang, J. Yang, Z. Zhang, B. Sikdar, H. Wang, H. Qiu, and M. Huang, "When Smiley Turns Hostile: Interpreting How Emojis Trigger LLMs' Toxicity," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 36, pp. 30431–30439, 2026.
- [11] M. Sriram et al., "Prompt Analysis in Large Language Models Adversarial Prompt Detection," *Research Digest on Engineering Management and Social Innovations*, vol. 2, no. 4, pp. 723–735, 2026.

- [12] R. Stasilovich et al., “Reprogramming LLM Semantics: A Symbolic Attack Using Emoji-Based Context Mutation,” 2026.
- [13] Y. Zhou, Y. Xiao, W. Ai, and G. Gao, “The Hidden Language of Harm: Examining the Role of Emojis in Harmful Online Communication and Content Moderation,” in *Proceedings of the Seventh Workshop on Natural Language Processing and Computational Social Science*, 2026.
- [14] H. Ashraf, H. M. M. Othman, and A. Mansour, “Towards Safer Large Language Models: A Comprehensive Survey of Harmful Content Detection, Mitigation, and Evaluation,” 2026.
- [15] S. Joo, H. Koh, and K. Jung, “Harmful Prompt Laundering: Jailbreaking LLMs with Abductive Styles and Symbolic Encoding,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 25489–25524, 2025.